

TUNG DUONG

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science; Specialization in Artificial Intelligence; GPA: 4.0/4.0

May 2026

Atlanta, GA

University of Massachusetts Boston

Bachelor of Science in Computer Science; Summa Cum Laude - Latin Honors; GPA: 3.92/4.0

May 2022

Boston, MA

SKILLS

AI & Machine Learning: Search Algorithms, Heuristic Modeling, K-Means Clustering, Exponential Smoothing, NLP.

Programming Languages: Python, SQL, JavaScript, TypeScript, Java, C++, C#.

Data & Architecture: Azure AI Search, FastAPI, Docker, D3.js, MySQL, REST APIs, JSON, React.

Eligibility status: I have authorization to work permanently in the United States.

EXPERIENCE

Applications Software Developer

University of Miami

Nov 2022 – Present

Coral Gables, FL (Remote)

- Architect backend API integrations and scalable data ingestion pipelines for enterprise platforms, utilizing Python and Azure AI Search protocols for advanced, large-scale metadata harvesting.
- Engineer highly secure, automated CI/CD deployment processes using Docker and Azure DevOps, optimizing container build environments by strictly separating build and runtime libraries.
- Lead comprehensive database refactoring initiatives, converting complex relational schemas to enforce data integrity, significantly reducing query latency across core institutional applications.
- Resolve high-severity vulnerabilities across distribution codebases and deploy Microsoft Entra ID (Azure SSO) to standardize and secure enterprise access protocols.

Office365 and System Administrator Apprenticeship

University of Massachusetts Boston

Aug 2020 – Jun 2022

Boston, MA

- Managed active directory lifecycles and large-scale data reconciliation as part of the Professional Apprenticeship and Career Experience (PACE) program.
- Automated enterprise data processing operations using scripting, significantly reducing manual ticket resolution time for large-scale communication pipelines.

PROJECTS [🔗 VIEW FULL PORTFOLIO HERE](#)

ARC-AGI Grid Transformation Agent | *Python, Machine Learning, Search Algorithms*

2025-2026

- Developed an AI agent that successfully solved 91% of complex grid transformation problems by discovering advanced transformation sequences to convert input meshes into target outputs.
- Integrated a custom transformation library with a beam search algorithm, a heuristic scoring system, and training set parameter inference to optimize dynamic logic paths.

U.S. Housing Market Analytics Platform | *Python, D3.js, Data Modeling, Clustering*

2025-2026

- Engineered a multi-scale, interactive web platform visualizing the impact of corporate investor activity on the U.S. housing market by integrating massive Zillow time-series and Redfin data sets.
- Implemented distance-based K-Means clustering to categorize city-level investor trends and utilized Holt-Winters exponential smoothing models to forecast future housing trajectories.

Patient-First Interoperable Portal | *Python (FastAPI), React, JSON, REST APIs*

2025-2026

- Architected a FHIR-compliant full-stack patient-provider portal that prioritizes patient agency and control over highly sensitive health data and medical charting.
- Engineered secure backend APIs allowing patients to selectively authorize and share specific subsets of data with providers via generated authentication codes.

AWARDS AND HONORS

Code4Lib 2023 Diversity Scholarship Award

Code4Lib Conference at Princeton University

2023

Princeton, NJ

Paul M. English '87 Computer Science Endowed Scholarship (\$10,000)

University of Massachusetts Boston

2021-2022

Boston, MA